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Concept Name: Texture in Images • Texture as a property distinguishing regions in an image. • • • • Texture is defined by patterns and arrangements of "things" (groups of pixels like dots, subdots or lines). Texture in computer vision refers to the visual patterns of surfaces, which helps in distinguishing different regions in an image not by color or intensity, but by their spatial arrangement. Imagine looking at a photo of a brick wall and a grassy field: although both might have similar overall colors, the brick wall has a repeating gridlike pattern (regular texture) while the grass is more random (irregular texture). Mathematically, texture is considered a function of both the "things" (pixel groupings forming identifiable marks) and the "patterns" (the arrangement of those things). Most natural textures are random or irregular, while regular or repeating patterns often indicate something man-made. • • • • The appearance of woven fabric (regular, repeating pattern) vs. a patch of gravel (random, irregular). • Figure 15.1 in the lecture shows various examples: dots (regular), lines (horizontal, vertical), patterns (tangled), noisy random textures (spray paint), or structured patterns (bricks, tiles). Concept Name, Name, Edge Detectors as Texture Operators, GPU Preferences, 2 Edge detectors highlight areas with rapid intensity change. • Textured regions have many edges; smooth regions have few. Edge detectors are originally designed to find boundaries (edges) between objects in images. Since textures often have many small variations, applying an edge detector reveals those variations. However, not all edge detectors are effective at distinguishing between different textures. Some, like Sobel, may detect many edges in both regular and random ways.

Textures, but may not differentiate them well because the output intensity (number of edges) is similar across different textures. • Figure 15.3 shows that the Sobel operator finds many edges in both tightly woven and random grass textures, but their outputs are not easy to distinguish. • Think of rubbing your finger over sandpaper (rough texture with many edges) vs. glass (smooth, few/no edges). Concept Name: Range Operator • Measures the difference between the maximum and minimum pixel values within a local neighborhood. The range operator evaluates texture by examining each small window (e.g., 3x3 pixels) and calculating the difference between the highest and lowest pixel values in that window. The resulting value is assigned to the center pixel. This highlights textured regions because, in areas with lots of variation, the difference between the brightest and darkest pixels is large. • Figure 15.4 shows how the range operator makes a patch of woven mat stand out from the rest of the carpet. • Imagine measuring the height difference between the tallest and shortest blades of grass in each small patch of a lawn: rough patches will give you bigger numbers. Concept Name: Variance Operator • Quantifies how much the pixel values deviate from each other in a local window. • Higher variance indicates more variability (i.e., more texture). Variance calculates how spread out pixel values are in a local window. For the custom definition (Equation 15.1), it sums the squares of the differences between the center pixel and the center pixel.

And each neighbor, then takes the square root. For the classical definition, it first computes the mean (average) of the window, then sums the squared differences between each pixel and the mean, divides by the window area, and (optionally) takes the square root to get the standard deviation. Formulas:• Substitution (custom):• Choreography:1.1.2.1,1,2.2,2,3,4,4.• Variance (custom: ∑(■■■■■■ ■■■■■■■■■■)2.3,1.4,2:1,3.11,4:2,5,6.1• Classical • • • • A noisy TV screen (static) has high variance, while a blank wall has low variance. Concept Name: Sigma (Standard Deviation) Operator: Sigma• Measures the spread (deviation) of pixel values within a small window. • Useful as a more interpretable measure than variance (since it's in original units). The sigma operator computes the standard deviation of pixel intensities in a small window. Standard deviation tells us how much individual pixel values usually differ from the mean value of the window. It's the square root of variance and provides a direct sense of intensity variability useful for segmentation. • Figure 15.6: The sigma operator distinguishes between a patch of mat and carpet. The color is high-but the contrast is low, needing histogram equalization (contrast enhancement) for better visualization. • Imagine measuring the height variation of pebbles within a small sandbox-stdlib.com-gives you an intuitive "average difference" from the mean height.

Concept Name: Skewness Operator • Measures the asymmetry of the histogram of pixel intensities in a window. • Helps identify areas where the distribution of intensities is not balanced around the mean. Skewness tells you whether the distribution of pixel values is symmetrical or "skewed" towards brighter or darker values. It's calculated as the mean of the cubed differences between each pixel and the mean, divided by the cube of the standard deviation (Equation 1.15.4). This is only useful in some textured regions where the intensity variation is non-asymmetrical. For a symmetric distribution (like a checkerboard), skewness is zero. Formula: ■■■■■1 1■■■■• ■■■ ■_■■■■■■■■= Â · d l ■3 ■∑(■■■■■■–■■■ Errors Figure 15.0.7.1.2.0x1.1x2. Skewness is zero for the checkerboard pattern (balanced black and white), but nonzero for random patterns (due to unbalanced distributions). • Think of throwing fair dice (symmetrical histogram, zero skewness) vs. loaded dice (skewed histogram, nonzero skewness). Comparison of Related Techniques • Sobel/edge detectors: Good for finding quick intensity changes, but not always for distinguishing textures. • •

• • Range, Variance, Sigma: Statistical operators, examine local variation-better-suited for quantifying texture differences. • Skewness: Goes beyond spread and looks at shape of pixel value distribution; it's unhelpful for detecting non-symmetric textural patterns.

Concept Name: Difference Operator • Measures the absolute difference in intensity between a pixel and another pixel a specified distance away. • • • • Fast method to highlight differences in texture patterns. The difference operator looks at the absolute difference between the intensity of one pixel and another pixel at a certain spatial offset. The size of this offset is crucial: if it matches the dominant "pattern size" of a texture, areas with repeating patterns will have low difference values (as similar pixels are sampled), while more irregular textures will have higher difference values. This operator runs through the image and, for each pixel, computes:■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ It runs faster than more complex statistics like variance or skewness, making it a practical tool for certain images. Errors Figure 15.8.8, 8.6.8x1. The woven texture gives low difference values (darker) while loose straw, which is woven,gives higher values (brighter) due to greater variability. • Think of comparing how similar tiles are along the length of a tiled floor: equally spaced tiles will have little difference, but random stones will show high differences. Concept Name: Mean Operator (Applied to Difference Array) • Smoothing or averaging the result of the difference operator. • Helps reduce noise and makes texture regions more uniform for segmentation. Once the difference operator has highlighted areas of texture change (possibly creating a noisy output), the mean operator (or local averaging) is applied. It smooths the result by replacing each pixel with the average value of its neighbors.

■■■■[■, ■] = (■/(■■■■■■■ ■■■■)) ∗
■■■(■■■■■■■■■■■■■■■■■■■■■■■■■) This reduces noise and sharpens the distinction between different texture regions, making segmentation (separating image regions) easier. Errors Figure 15.9.9x1.0x2.1x11. Smoothing the output of the difference operator produces fuzzier (less-noisy) but more uniform regions for each texture. • Think of blurring a noisy photograph: local variations are suppressed, but the overall pattern becomes clearer. Concept Name: Hurst Operator• Multi-scale measure of how pixel value ranges change as you look farther from the center-to-center pixel. • Uses log-log slope of range-vs-distance plot for texture characterization. The Hurst operator investigates texture at multiple scales. For a given window (e.g., 7x7), it considers concentric "rings" (groups of pixels at the same distance from the center). For each ring, it computes the range (max-min pixel value). By plotting the natural logarithm (ln) of the ranges against the ln of ring distances, and fitting a straight line to these points, the slope of the line becomes the texture measure: low slopes for

smooth textures, high slopes for rough textures. This exploits how rough, complex textures exhibit greater variability as you 'look further away'. 1. For each ring/class of pixels at distance r from the center, compute range = r.max(ring) - min(ring). 2. For all rings (various distances), plot ln(range) vs ln(distance). 3. Fit a straight line; its slope is the Hurst measure. • Figures 15.11 & 15.12: A smooth image section yields slope ~0.99, a rough one yields ~1.0, and a smooth oneyields slope ~2.0.

Errors Figure 15.0.13.13,13.1.1 shows how the operator can segment smooth from rough regions in an embarrassingly artificial example. • Analogy: Like examining mountain ranges-the further you move from the center of the mountain to the center-a peak, the greater the difference in elevation if the terrain is rugged (high slope), but gentle hills yield low differences (low slope). Does not always work in real, complex images. Figure 15.14 shows it failing to properly segment textures in a house image, acting more like an edge detector. Concept Name: Compare Operator • Matches a sample texture patch against the entire image via pixel-wise comparison. • Useful for finding regions similar to a reference texture. The compare operator slides a small sample patch (e.g., a 3x3 or 5x5 region representing a known texture) across the whole image. For each position, it calculates the absolute difference between the sample and the underlying window in the image, and uses the average absolute difference as the output. Low values indicate similar textures, and high values indicate dissimilar textures. This approach is intuitive and works well for finding one texture of interest in an image. 1. Choose/copy a "template" or sample patch from the image (e.g., patch of brick). Ance 2.0. 1. Slide this patch over the image. 3. For each position, compute the mean absolute difference between the patch and the underlying area. 4. Output is the mean difference: small output means the area matches the sample size/texture. • Figure 15.15: The operator distinguishes between a woven and straw texture using a sample from the woven area.

• Figure 15.16: Using a brick patch on a house image, bricks are darkest (smaller) outputs (difference), while trees/roof/windows have higher (brighter) outputs. • Analogy: Like holding a fabric swatch against various clothes in a store-if the colors/colors/patterns match, the difference is small and you know it's the same fabric. Can only highlight one texture at a time. Other textures become indistinguishable, being lumped together as not matching. "The Difference Operator"Additional Notes and Comparison:• Difference Operator vs. Classical Edge Detectors: Both highlight changes, but the difference operator can be tuned to a particular spatial scale, making it adaptive to texture size. • Mean Operator acts as a post-processing smoothing step, reducing noise and making class boundaries more distinct for segmentation. • Hurst Operator: More complex, multi-scale analysis; gives a single number per region/region; computationally heavy, not universally reliable.• Compare Operator:

Powerful for template-based matching, but inherently limited in its ability to handle multiple textures simultaneously. Summary: Each operator is best suited for particular texture recognition/segmentation tasks. In practice, combining several operators or preprocessing/post-processing steps (like smoothing or histogram equalization) can provide the most robust texture analysis in images.